P1-07 — Decision Dossier for PC-1 … PC-6 (No Automatic Approval)

Command: P1-07 — PC-1 to PC-6 Decision Dossier **Date:** 21 October 2026 **Purpose:** analysis and documentation only. **Nothing in this dossier approves, rejects, or integrates any proposal.** Every status below remains **PENDING — Boss/Commander decision required**.

Sources cross-checked: 02_THEOREM_BLUEPRINT.md (frozen text, transcribed), 04_BLUEPRINT_GAP_REGISTER.md (GAP-01, GAP-06, GAP-17, outcomes of P1-03…P1-06), P1_03_MATHEMATICAL_PROOF_PLAN.md (§4, §6), P1_04_PROOFS_T1_T4.md, P1_05_PROOFS_T5_T6_COROLLARIES.md, P1_06_MOMENT_PROOFS_C5_C6.md, and the validation scripts/outputs code/validate_p1_03.py, code/validate_p1_05.py, code/validate_p1_06.py (all rerun unmodified for this dossier; §8).

How to use this dossier

For each PC: (1) exact frozen text affected, (2) exact proposed old→new change, (3) mathematical reason, (4) proof location and evidence, (5) classification, (6) consequences of accepting/rejecting, (7) **recommended disposition (recommendation only)**. The Boss/Commander should record, for each PC, one of APPROVE / REJECT / DEFER plus date; integration (if any) must happen in a later command as dated correction notes, never silently (GAP-17 resolution rule).

Summary table (recommendations only — all statuses PENDING)

PC | Affects | Class | Recommendation

PC-1 | C5 hull mean | **Theorem correctness** (frozen formula false) | **APPROVE**

PC-2 | T3 / convention freeze | Convention consistency (correctness-critical ambiguity) | **APPROVE**

PC-3 | T2 / definition of # | Correctness of quantification (definition domain) | **APPROVE**

PC-4 | C4 (add dual-containing count) | Exposition/completeness (proved addition, no change to existing text) | **APPROVE**

PC-5 | T1 / global scope | Scope/exposition | **APPROVE**

PC-6 | C6 status marking | Exposition/status accuracy | **REJECT as superseded** (RR-8 has since been proved; replace instead by integrating the P1-06 formulas — a separate integration action, not PC-6)

PC-1 — C5 hull mean: split L = 1 / L ≥ 2

1. Exact frozen statement affected (02_THEOREM_BLUEPRINT.md, C5): “E[Hull] = [P(2P+1) / (6(P+1))] · Σ_O w_O L_O”, with proof key: “ek kenare par min(P−α_i, α_{i−1}) do swatantra uniform {0,...,P} ka min hai; E[min] = Σ_{t=1}^P ((P−t+1)/(P+1))² = P(2P+1)/(6(P+1)). P=1 par 1/4 — purana mean wapas.” (The first part of C5, E[dim C] = (n/2)·(Σ_s m_s), is correct and is **not** touched by PC-1; re-derived in P1-06 §2.1.)

2. Exact proposed change. Replace the displayed formula and key by: E[Hull] = μ_P · Σ_{s} Σ_{O∈O_s, L_O ≥ 2} w_O·L_O + ν_P · Σ_{s} Σ_{O∈O_s, L_O = 1} w_O, where μ_P = P(2P+1)/(6(P+1)) and ν_P = ⌊P²/4⌋/(P+1). Replace the “P=1 par 1/4” sentence by: “P=1 par E[Hull] = (1/4)·Σ_{L_O ≥ 2} w_O L_O (L_O = 1 orbits ka yogdaan 0 hai).”

3. Mathematical reason. The proof key is valid only when an edge’s two arguments come from distinct independent exponents. For L_O ≥ 2 that holds (each cyclic edge pairs A_i with the distinct A_{i−1}; also for L_O = 2, the two edges are the two ordered pairings of the same distinct pair), so every edge has mean μ_P (P1-06 Lemma 1.1, tail-sum derivation) and the orbit mean is L_O·μ_P (Prop. 2.3). For L_O = 1 the single edge is a self-loop min(P−A, A) whose arguments are functions of one variable; its mean is ν_P (Prop. 2.2, parity evaluation of Σ_a min(P−a,a) = ⌊P²/4⌋). Since ν_P ≠ μ_P for all P ≥ 1 (e.g. ν_1 = 0 ≠ 1/4 = μ_1), the frozen display overcounts exactly the L_O = 1 part.

4. Proof location and evidence. Proof: P1_06_MOMENT_PROOFS_C5_C6.md §2.2–2.4 (Theorem 2.4), building on P1_03_MATHEMATICAL_PROOF_PLAN.md §4. Exact counterexamples to the frozen display (vector-level Gram-rank brute force, P1_03 §7, GAP-01): GF(2), n_0=7, P=2: brute mean 11/3 vs frozen 35/9; GF(3)/GF(9), λ=1, P=3 (four L=1 orbits): brute 2 vs frozen 7/2; GF(8), k=1, P=2: brute

11/3 vs frozen 35/9; in L≥2-only cases (λ=−1) frozen = corrected = brute (7/2), isolating the error to L=1. Orbit-level: validate_p1_06.py confirms ν_P, μ_P against enumeration for P ∈ {1,2,3,4,5,8,9} (ALL P1-06 MOMENT CHECKS PASSED, rerun for this dossier).

P = 1 special-case verification (required). At P = 1: ν_1 = ⌊1/4⌋/2 = 0 (indeed min(1−A, A) ≡ 0 on A ∈ {0,1}), μ_1 = 1·3/(6·2) = 1/4. Correct prediction: an L = 1 orbit contributes 0; each L ≥ 2 orbit contributes w_O·L_O/4. Frozen prediction: w_O·L_O/4 also for L = 1 orbits — wrong by w_O/4 per unit-weight L = 1 orbit. Example: the GF(3), λ=1, P=3 case has four L=1, w=1 orbits: correct contribution 4·ν_3 = 4·(1/2) = 2 (brute 2), frozen 4·1·μ_3 = 4·(7/8) = 7/2. The cleanest P=1 instance is any single L=1, w=1 orbit: true mean 0, frozen prediction 1/4.

5. Classification. Theorem correctness. This is the only PC that repairs a false theorem statement.

6. Consequences. Accept: C5 becomes true; the P=1 recovery note for moments becomes correct; P1-06 Theorem 2.4 can be cited as the proof. Nothing else in T1–T6/C1–C4 changes (the enumerator never used the mean). Reject: the blueprint permanently carries a false formula; any manuscript must either omit C5's hull mean or flag it as an open error — a serious credibility risk, since the counterexamples are elementary (any L=1 orbit). Defer: same as reject until decided.

7. Recommended disposition: APPROVE — the frozen formula is demonstrably false and the replacement is fully proved and multiply verified.

PC-2 — Add the dual-order sentence to the convention freeze

1. Exact frozen text affected (02_THEOREM_BLUEPRINT.md §0, freeze): "Code-first pairing: ⟨x,y⟩_{s,k} = Σ_i x_i σ(y_i); dual D(C), hull Hull_k(C) = C ∩ D(C)." The symbol D(C) is used (here and in T3) **without a defining set-builder expression**; the word "code-first" in the header ("Convention freeze: … code-first pairing, inverse-Frobenius reciprocal") is the only indication of the intended order.

2. Exact proposed change. Add one sentence to the freeze: "D(C) = {y : ⟨c, y⟩_{s,k} = 0 for all c ∈ C} (code vector first, candidate second)." No other wording changes.

3. Mathematical reason — trace through RR-2, RR-3, RR-4 (with the roles of σ, ρ, h^♮, h_a#).

- RR-2 (#, τ): # is defined by coefficientwise ρ plus reversal (f#(x) = ρ(f_0)^{-1} Σ ρ(f_i) x^{r−i}); it is multiplicative and permutes the factor set F. This is pure algebra — the dual-order convention plays no role here, but # is the object whose correctness as a ∗dual generator∗ depends on the order, next steps.
- RR-3 (Euclidean dual): the Euclidean pairing Σ c_i z_i is symmetric, so the Euclidean dual needs no order convention; its generator is h^♮, the **ordinary** reciprocal of h_a (no Frobenius), proved in P1-04 Lemma RR-3 without separability.
- RR-4 (transport): the code-first dual solves ⟨c, y⟩ = Σ_i c_i σ(y_i) = 0 for the **second-slot** candidate y: setting w = σ(y) shows y ∈ D(C) ⟺ σ(y) ∈ C^⊥ ⟺ y ∈ ρ(C^⊥) (coordinatewise ρ = σ^{-1}); applying the coefficientwise ρ to h^♮ produces exactly h_a#, with exponents β_i = P − α_{i−1} (P1-04 Prop. 4.2, T3). Under the opposite slot order, the same computation gives D(C) = σ(C^⊥), the generator built from coefficientwise σ instead of ρ, and predecessor indices reversed (β_i = P − α_{i+1} under the same orbit labeling) — i.e. the frozen h_a#/exponent formula would be **wrong**, not just differently worded. The freeze as written does not exclude that reading on paper.

4. Proof location and evidence. P1_04_PROOFS_T1_T4.md (S7) states the sentence verbatim and uses it in Prop. 4.2; P1_05_PROOFS_T5_T6_COROLLARIES.md §1/G5 rechecks it. Computational evidence that the intended reading is code-first: the GF(8), k=1 case (σ = a² ≠ a⁴ = ρ) in validate_p1_03.py matches the ρ-formula (2,187 assignments; the old GF(4) prototype cannot distinguish the readings since there σ = ρ). Register: GAP-17.

5. Classification. Convention consistency with direct correctness consequences (an ambiguous

freeze makes T3's statement reader-dependent).

6. Consequences. Accept: T3/T4 become unambiguous; the sentence already used in P1-04/P1-05 becomes the frozen one; no theorem content changes. Reject: every future proof/manuscript must restate the convention locally (as P1-04/P1-05 did) and a referee can legitimately read T3 the other way. Defer: workable short-term (proof files carry the sentence), but leaves the root ambiguity in the frozen source.

7. Recommended disposition: APPROVE — one sentence, zero mathematical change to the intended (and computationally confirmed) reading, removes a genuine ambiguity.

PC-3 — Add the hypothesis f(0) ≠ 0 to the range of

1. Exact frozen text affected (02_THEOREM_BLUEPRINT.md §0 and T2): the reciprocal is introduced as "f#(x) = ρ(f₀)⁻¹ Σ_i ρ(f_i) x^{r−i} (monic, degree-rakh rakhta hai)", and T2 states "(i) # monic polynomials par multiplicative hai; … (ii) Root action: f(α)=0 ⟹ f#(α^{−p^{d_s−k}})=0", without restricting which monic polynomials # applies to.

2. Exact proposed change. Wherever # is defined/quantified, read: "for monic f **with f(0) ≠ 0**". Proposed sentence for the freeze: "# is defined for monic polynomials with nonzero constant term."

3. Mathematical reason. The definition divides by ρ(f_0); if f(0) = f_0 = 0 this is division by zero — # is undefined, not merely inconvenient. Minimal witness: f(x) = x (monic) has no reciprocal of the displayed form. The hypothesis is automatic in every application in the paper (factors f ∈ F_s divide S_s and S_s(0) = −μ_s ≠ 0; likewise h_a(0) ≠ 0 since g_a(0)h_a(0) = −λ_s), so adding it narrows nothing that is ever used; it only makes T2's quantifier honest.

4. Proof location and evidence. P1_04_PROOFS_T1_T4.md (S6), Lemma RR-2(ii)/(iv) (the root-transport proof uses α ≠ 0, i.e. f(0) ≠ 0), and the §7 hypothesis audit ("# undefined when the constant term vanishes; such factors never divide S"). Register: P1-03 §6 review of T2.

5. Classification. Theorem correctness (well-posedness of a definition/quantifier); effectively a domain clarification.

6. Consequences. Accept: T2 is literally true as written; no proof step changes (P1-04 already assumed it). Reject: the frozen T2 remains formally ill-posed for e.g. f(x) = x — a trivial but real referee objection. Defer: same exposure until decided.

7. Recommended disposition: APPROVE — necessary for literal correctness, costless in scope.

PC-4 — Add the dual-containing count to C4

1. Exact frozen statement affected (02_THEOREM_BLUEPRINT.md, C4): "C ⊆ D(C) ⟺ har orbit-edge par α_{i−1}+α_i ≥ P; ginati ∏_O tr(B_P^{L_O}), (B_P)_{a,b} = 1 agar a+b ≥ P varna 0. Ulta: D(C) ⊆ C ⟺ α_{i−1}+α_i ≤ P har edge par." The characterization of D(C) ⊆ C is stated; **no count** is stated for that side.

2. Exact proposed change. Append to C4: "#{a : D(C) ⊆ C} = ∏_O tr((B′_P)^{L_O}), (B′_P)_{a,b} = 1 agar a + b ≤ P varna 0." Nothing else changes.

3. Mathematical reason. The frozen characterization says dual-containing assignments are exactly the closed walks whose steps satisfy a + b ≤ P; by the same trace expansion as T5 (P1-05 §2, Step 1, with the 0/1-matrix B′_P), their number per orbit is tr((B′_P)^{L_O}), and independence over orbits gives the product. Moreover the two counts in C4 coincide: α_i ↦ P − α_i maps {a+b ≥ P} walks bijectively to {a+b ≤ P} walks.

4. Proof location and evidence. Proved as the labeled PC-4 proposal in P1_05_PROOFS_T5_T6_COROLLARIES.md §5 (PC-4 box, including the involution). Computational: validate_p1_03.py checks the dual-containing brute counts against ∏ tr((B′)^L) in all six vector cases (e.g. GF(9), λ=−1: 100 = 100; GF(8): 258 = 258); validate_p1_05.py confirms the global count 144 = 144 in the two-component case. Both rerun for this dossier (§8).

5. Classification. Exposition/completeness: a proved addition; no existing frozen claim is altered.

6. Consequences. Accept: C4 becomes symmetric and complete; the manuscript can cite the count as a corollary. Reject: the count remains available only as the P1-05 proposal; the characterization (the mathematically essential part) is unaffected. Defer: harmless; can be decided together with manuscript drafting.

7. Recommended disposition: APPROVE — fully proved and verified; pure completion. (If the Boss prefers a minimal-diff philosophy for C4, REJECT is defensible and costs only a citation label.)

PC-5 — State in T1 that global codes are A-submodules with fixed labels

1. Exact frozen statement affected (02_THEOREM_BLUEPRINT.md, T1): “Har component code C_s ⊆ K_s[x]/(M_s) ka ekmatra monic generator M_s ka divisor hai, isliye a ↦ C_s(a) ek bijection hai … CRT se global codes A-par ⟷ assignments ka product. Kul ginati (P+1)^{Σ_s |F_s|}.” The component-level statement is complete; the global step invokes “global codes” without saying what a code over A is.

2. Exact proposed change. Add to T1 (or to §0’s global model): “Global code se arth hai A^n ka A-submodule (fixed labeled CRT decomposition A ≅ ∏_s K_s ke tahat); component codes K_s-linear hain, aur ginati labeled assignments ki hai, equivalence classes ki nahi.” (English gloss: a code over A is an A-submodule of A^n under the fixed labeled decomposition; counts are over labeled assignments.)

3. Mathematical reason. The product parametrization is a statement about A-submodules: an A-submodule of A^n ≅ ∏_s K_s^n decomposes as a product of K_s-subspaces (P1-05 Prop. 3.1(ii), via the component idempotents of A). For arbitrary F_q-linear subspaces of the underlying F_q-space, this is false in general — a subspace mixing different components need not be a product — so without the A-submodule clause, T1’s global bijection has a genuine scope hole. The project’s own scope lock already intends this (04_PAPER_BLUEPRINT.md §3, locks 1–2: “A-submodule … component labels and the CRT isomorphism are fixed … never over equivalence/isometry classes”), but the frozen theorem text does not carry it.

4. Proof location and evidence. P1_05_PROOFS_T5_T6_COROLLARIES.md (G3), Prop. 3.1 (assembly proved under exactly this hypothesis), §0 of P1_06_MOMENT_PROOFS_C5_C6.md (uniform labeled model). No counterexample is involved; this is a scope-precision issue. Register: P1-03 §6 review of T1 (“PC-5, wording only”), 04_PAPER_BLUEPRINT.md §3.

5. Classification. Scope/exposition (the proofs already work under the intended definition).

6. Consequences. Accept: T1/T6’s counts are unambiguously about the labeled A-submodule model that all proofs and validations use. Reject: the global statement stays open to a mixing-subspaces misreading; the defense would have to cite the paper blueprint instead of the theorem file. Defer: tolerable internally, risky at manuscript stage.

7. Recommended disposition: APPROVE — aligns the frozen theorem with the model that was actually proved and computed.

PC-6 — Mark C6 “to be proved (RR-8)” — now superseded

1. Exact frozen statement affected (02_THEOREM_BLUEPRINT.md, C6): “Variance: Orbit-swatantra yog; L=1 …, L=2 …, L≥3 … — teeno cases ke band sutra (closed forms) Phase 4 me P ke parimaani (polynomial) roop me nikale jayenge. **[RR-8]**” At the time PC-6 was registered (P1-03), no closed forms existed, and GAP-06 required C6 to be marked as unproved wherever summarized, lest a summary overclaim a variance theorem.

2. Exact proposed change (as registered). Add the marking “to be proved (RR-8)” to C6 and to any summary of it, until RR-8 is discharged.

3. Mathematical reason — and why the situation changed. The marking was a truth-in-labeling device during P1-03…P1-05. Since then, P1_06_MOMENT_PROOFS_C5_C6.md §3 (Theorem 3.7) proves the closed forms: parity-split Var(S_1); Var(S_2) = $P(P+2)(P^2+2P+3)/(18(P+1)^2)$; Var(S_L) = $L\cdot P(P+2)(P^2+2P+2)/(60(P+1)^2)$ for $L \geq 3$; globally $\Sigma_{s,O}\ w_O^2\ \mathrm{Var}(S_{L_O})$. RR-8 is therefore discharged mathematically, and a "to be proved" label would now be **false**.

4. Proof location and evidence. P1_06_MOMENT_PROOFS_C5_C6.md §§1, 3 (Lemmas 1.2, 3.2, 3.3, 3.5; Props. 3.1, 3.4, 3.6); code/validate_p1_06.py exact-rational verification (rerun, §8). Register: GAP-06 and the P1-06 outcome.

5. Classification. Exposition/status accuracy.

6. Consequences. Accept (as worded): the blueprint would assert the variance is unproved — now incorrect; would need immediate re-amendment after integration. Reject: C6's frozen program sentence ("closed forms will be derived in Phase 4") temporarily remains, which is now an understatement rather than an overclaim; the accurate fix is a different edit (below). Defer: leaves a stale status label question open with no benefit.

7. Recommended disposition: REJECT as superseded. The protective intent of PC-6 is fully honored by recording here (and in GAP-06) that RR-8 was open until P1-06 and is now proved. The correct follow-up edit — replacing C6's program sentence by the P1-06 Theorem 3.7 formulas — is **not** PC-6 as registered and must be proposed/decided as part of the integration command, e.g.: "C6 Variance: $\mathrm{Var}(\dim_{F_q}\mathrm{Hull}_k) = \Sigma_{s,O}\ w_O^2\ V_{L_O}(P)$ with V_1, V_2, V_L ($L\geq 3$) as in P1-06 Theorem 3.7."

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7. Interaction note

PC-1 and the C6 replacement suggested under PC-6 are siblings: both repair blueprint corollaries whose mathematics is now settled (P1-06). Approving PC-1 without planning the C6-formula integration would leave the blueprint half-updated; the recommended sequencing is a **single** future approval/integration command deciding PC-1…PC-6 (this dossier's recommendations: approve PC-1…PC-5, reject PC-6-as-worded while adopting the C6 formula replacement), applying all accepted changes as dated correction notes in one edit, and recording the Git hash. Until that command, **all six statuses remain PENDING and the frozen file remains untouched.**

8. Checks run for this dossier (unmodified scripts)

```
python3 code/validate_p1_03.py   # exit 0; output byte-identical to code/validation_p1_03_output.txt; ALL P1-03 EXTENDED CHECKS PASSED
python3 code/validate_p1_05.py   # ALL P1-05 MULTI-COMPONENT ASSEMBLY CHECKS PASSED
python3 code/validate_p1_06.py   # ALL P1-06 MOMENT CHECKS PASSED
```

These reruns support the evidence columns above (PC-1 means, PC-2 $\sigma \neq \rho$ placement, PC-4 counts, C6 formulas). They are finite computational evidence; the proofs cited are the actual warrants. No earlier proof or validation artifact was modified; 02_THEOREM_BLUEPRINT.md is untouched (git diff empty), and **Gate G1 (novelty) remains open and untouched by this dossier** — none of PC-1…PC-6 concerns literature priority.